

Smart Recommendation System– Project 5

AI/ML Internship Project Report

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Recommendation systems are an important application of Artificial Intelligence and Machine Learning. They help users discover items that match their interests by analyzing similarities between products or content.

This project recommends similar movies based on the genre of a movie selected by the user. It uses a content-based filtering approach along with Cosine Similarity to generate recommendations.

Objectives

- Learn recommendation system concepts.
- Implement content-based filtering.
- Generate personalized movie recommendations.
- Apply Machine Learning similarity techniques.
- Improve user experience through intelligent suggestions.

Tools & Technologies

Programming Language

- Python

Libraries

- Pandas
- Scikit-learn

Machine Learning Components

- CountVectorizer
- Cosine Similarity

Platform

- Google Colab

Dataset

The dataset contains the following attributes:

Movie	Genre
Avengers	Action Sci-Fi
Iron Man	Action Sci-Fi
Batman	Action
Superman	Action
Titanic	Romance Drama
Avatar	Sci-Fi Adventure
Inception	Sci-Fi Thriller

Interstellar	Sci-Fi Adventure
The Dark Knight	Action Thriller
Doctor Strange	Action Fantasy

The data is stored in a CSV file named **movies.csv**.

The project follows these steps:

1. Load the movie dataset.
2. Convert movie genres into numerical features using **CountVectorizer**.
3. Calculate similarity scores using **Cosine Similarity**.
4. Accept a movie name as user input.
5. Find movies with the highest similarity score.
6. Display the top recommended movies.

Features

- Personalized Movie Recommendation
- User Preference Input
- Content-Based Filtering
- Cosine Similarity Scoring
- CSV File Handling
- Fast Recommendation System

This project uses **Content-Based Filtering**.

Input

- Movie Name

Processing

- Genre Vectorization using CountVectorizer
- Similarity Calculation using Cosine Similarity

Output

- List of Recommended Movies

The recommendation engine compares movie genres and suggests movies with similar characteristics.

Result

The recommendation system successfully suggests similar movies based on the selected movie.

Sample Input

Movie Name:

Avengers

Sample Output

Recommended Movies:

Iron Man

Doctor Strange

Batman

The recommendations are generated based on genre similarity using Cosine Similarity.

Future Scope

- Expand the movie dataset with more titles.
- Add movie ratings and user reviews.
- Develop a graphical user interface (GUI).
- Support recommendations for books, music, and TV shows.
- Implement hybrid recommendation techniques combining content-based and collaborative filtering.

The Smart Recommendation System demonstrates the practical use of Machine Learning in building personalized recommendation engines. By applying CountVectorizer and Cosine Similarity, the system effectively recommends movies based on user preferences. This project highlights the importance of recommendation systems in modern digital platforms and provides a foundation for developing more advanced AI-powered recommendation solutions.